

Clinical Risk Assessment Report: Patient 18

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.52x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.1%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.68 [Bottom 23%] (-3.1%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.59 [Bottom 24%] (-0.9%)**
Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.37 [Avg (63th %ile)] (-0.5%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.11 [Avg (51th %ile)] (-0.5%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.6 [Avg (27th %ile)] (-0.4%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.35 [Avg (53th %ile)] (-0.4%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.81 [Avg (46th %ile)] (-0.3%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: -2.11 [Bottom 5%] (-0.2%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

DISCLAIMER: This report is generated by an investigational machine learning model. It is intended to provide adjunctive biological insights and must not replace standard clinical, radiological, or pathological evaluation.

Machine Learning Feature Attribution (SHAP Decision Path)

